

Vertex Formula



- 1 The x -coordinate of the vertex of the parabola occurs at $x = \frac{-b}{2a}$. The y -coordinate of the vertex is found by substituting $x = \frac{-b}{2a}$ into the parabola's equation and evaluating the function at this value of x .
- 2 Axis of symmetry
- 3
 - a $x = 0, 6$
 - b $(3, 9)$
- 4
 - a Concave up
 - b $x = 9$
 - c $y = -71$
- 5
 - a Concave up
 - b $x = 1, 5$
 - c $y = 5$
 - d $x = 3$
 - e $(3, -4)$
- 6
 - a Concave down
 - b $x = -4, 2$
 - c $y = 8$
 - d $x = -1$
 - e $(-1, 9)$
- 7
 - a Concave down
 - b $x = -4, -2$
 - c $y = -8$
 - d $x = -3$
 - e $(-3, 1)$
- 8
 - a $x = -2$
 - b -1
 - c $(-2, -1)$
 - d $(-2, -2)$
 - e
 - i Yes
 - ii No
 - iii Yes

9

a $x = 14,6$

b $x = 10$



c Concave down

d $y = 16$

10 $(1, -12)$

11 $y = 0$

12 a -0.087

b -1.24

13 $x = \frac{1}{4}$

14 a Vertex of $P_1 = (-2, 2)$
Vertex of $P_2 = (2, 2)$

b 4

15 a $x = -3$

b -5

Applications

16 a $s = 12$

b 337

17 a $t = 3$

b 122 units